

# MESSIER 12 GLOBULAR CLUSTER

Sanjeev K (PH25MSCST11026)

Indian Institute of Technology Hyderabad

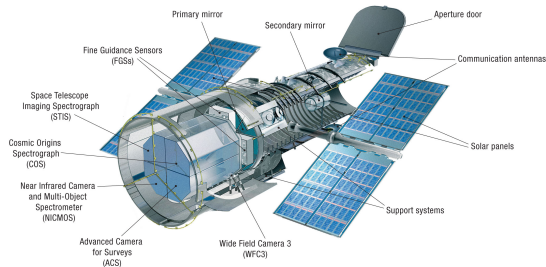
## 1 Introduction to Hubble and Messier 12

The Hubble Space Telescope (HST) is a 2.4-meter reflecting telescope deployed in low-Earth orbit at 600 km altitude on April 25, 1990, by the crew of Space Shuttle Discovery (STS-31). It was designed as a long-term space-based observatory with regular servicing missions. HST carries three cameras (ACS, WFC3/UVIS, WFC3/IR), two spectrographs (COS and STIS), and Fine Guidance Sensors (FGS). Since it operates above the atmosphere, it can observe ultraviolet light (blocked by Earth's atmosphere) and infrared without atmospheric background noise — giving much cleaner images than ground-based telescopes. In total, HST underwent five servicing missions (1993, 1997, 1999, 2002, 2009), each upgrading instruments and repairing equipment.

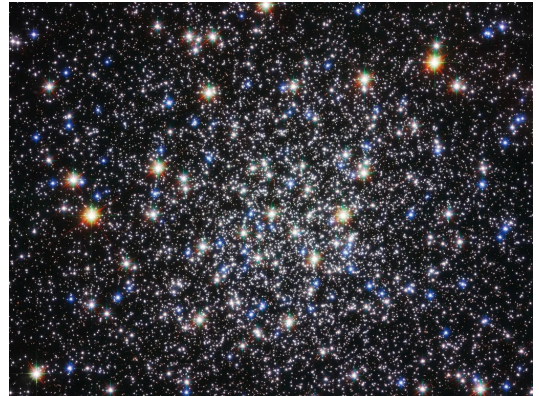
Messier 12 (M12) is a globular cluster discovered by Charles Messier in 1764. Located about 23,000 light-years away in the constellation Ophiuchus, M12 has an apparent magnitude of 7.7 and is best observed during July using binoculars.

Like many globular clusters, M12 contains binary star systems in which two stars orbit each other. In some cases, matter transfers from one star to its companion, producing X-rays that help astronomers identify these systems. Interestingly, M12 contains many such X-ray sources despite being a relatively diffuse cluster where stellar interactions are expected to be less frequent.

Astronomers have also discovered that M12 contains fewer low-mass stars than expected. It is believed that gravitational interactions with dense regions of the Milky Way stripped away nearly one million stars during the cluster's orbit around the galaxy. Hubble observed the center of M12 using both visible and infrared imaging techniques.



(a) Hubble Space Telescope



(b) Globular Cluster M12

Figure 1: *Left: The Hubble Space Telescope (Image source: Mikulski Archive for Space Telescopes). Right: The globular cluster M12 captured by the Hubble Space Telescope (Image source: NASA).*

## 2 Procedure

The complete workflow followed in this project is shown in Figure 2. The analysis pipeline consists of FITS image loading, visualization, background estimation, stellar source detection, and flux analysis.

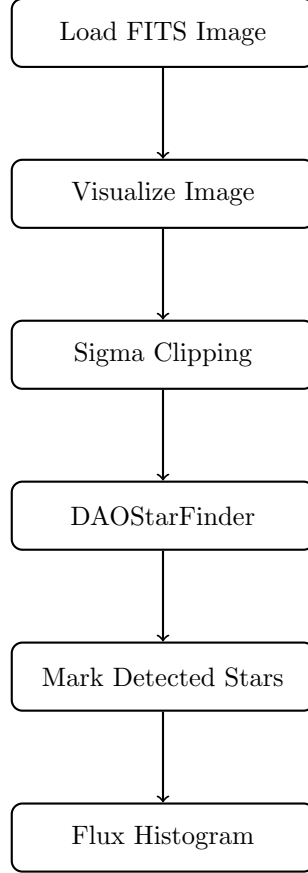


Figure 2: Workflow of the Messier 12 analysis pipeline.

### 2.1 Loading the FITS File

The astronomical image stored in the `vcomb.fits` file was loaded using the `astropy.io.fits` module. The science image was extracted from the primary HDU as a  $2048 \times 2048$  floating-point array for further analysis.

### 2.2 Image Visualization

The FITS image was visualized using `matplotlib.pyplot.imshow()` with the `inferno` colormap. Since astronomical images contain a large dynamic range, logarithmic normalization using `LogNorm()` was applied to enhance faint stellar structures while preserving bright regions.

### 2.3 Background Estimation using Sigma Clipping

The background statistics of the image were estimated using the Astropy function `sigma_clipped_stats()`. Sigma clipping iteratively removes outlier pixels beyond a specified threshold, thereby reducing the influence of bright stars during background estimation.

A clipping threshold of  $\sigma = 3.0$  was used to compute the background mean, median, and standard deviation values.

## 2.4 Star Detection using DAOSTarFinder

Stellar sources were detected using the `DAOSTarFinder` algorithm from the Photutils package. The detector was initialized with a Full Width at Half Maximum (FWHM) value of 3.0 pixels and a detection threshold equal to five times the estimated background standard deviation. Before source extraction, the median background value was subtracted from the image to suppress background contributions and improve detection accuracy.

## 2.5 Marking Detected Stars

The centroid positions returned by `DAOSTarFinder` were used to overlay circular apertures on the logarithmically scaled image using `CircularAperture`. The detected stars were marked with green circular boundaries for visualization of stellar positions across the cluster field.

## 2.6 Flux Analysis

The flux values of the detected stars were extracted from the source catalog generated by `DAOSTarFinder`. A logarithmic flux histogram using  $\log_{10}(\text{Flux})$  was then plotted to study the brightness distribution of stars in Messier 12. The histogram revealed that the majority of stars possess relatively low flux values, while only a small fraction of stars exhibit extremely high brightness.

# 3 Results and Discussion

The FITS image of Messier 12 was successfully processed and analyzed using Python-based astronomical data analysis tools. The complete pipeline included image visualization, background estimation using Sigma Clipping, stellar source detection using `DAOSTarFinder`, and flux analysis.

Figure 3 shows the original FITS image displayed with logarithmic normalization. The use of `LogNorm()` improved the visibility of faint stellar structures across the globular cluster field.

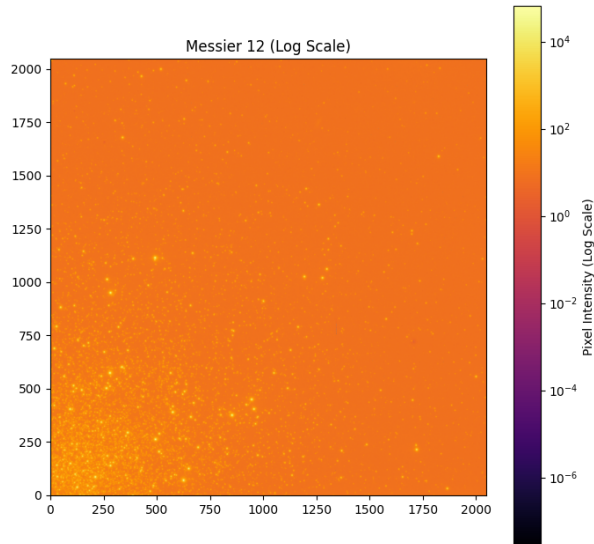


Figure 3: Log-scaled visualization of the Messier 12 FITS image.

Background statistics estimated using `sigma_clipped_stats()` provided stable mean, median, and standard deviation values by reducing the influence of bright stellar outliers. These values were then used to define a reliable detection threshold.

Figure 4 shows the stars detected by **DA0StarFinder** and marked with circular apertures. The source extraction successfully identified stellar objects across the dense cluster region.

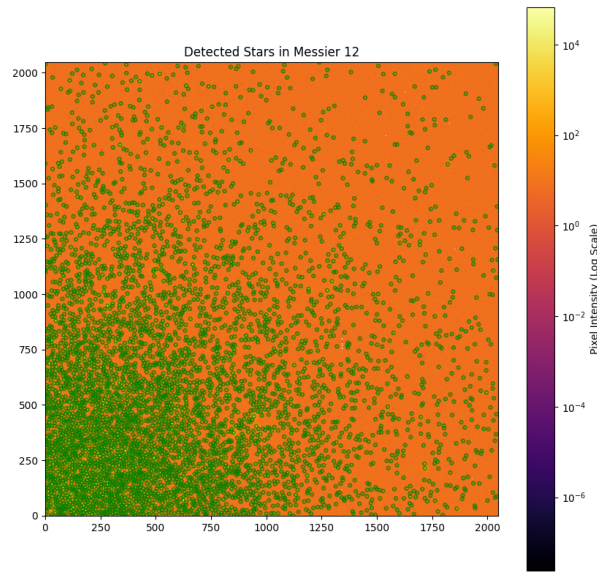


Figure 4: Detected stars in Messier 12 marked using circular apertures.

The flux values of the detected stars were analyzed using a histogram. Figure 5 shows the logarithmic flux distribution of the detected stellar population. The distribution indicates that most stars are faint, while only a small fraction exhibit very high flux values, which is typical of globular cluster systems.

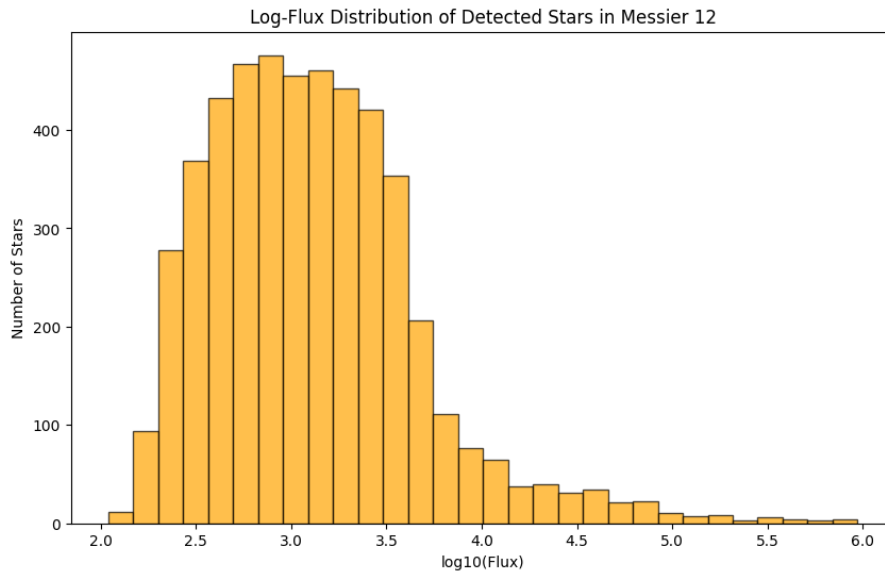


Figure 5: Logarithmic flux distribution of detected stars in Messier 12.

Overall, the project demonstrates a complete astronomical image-analysis workflow involving FITS data handling, background estimation, stellar source detection, and photometric analysis.

## 4 Python Code

### 4.1 Data Loading and Visualization

```
#importing the required libraries
import numpy as np
import matplotlib.pyplot as plt
from astropy.io import fits
from astropy.stats import sigma_clipped_stats

# Loading the FITS file
file_path = '/home/Interstellar/Desktop/M12_Analysis/data/Vcomb.fits'
hdul = fits.open(file_path)
header = hdul[0].header
data = hdul[0].data
hdul.close()

# Printing basic information about the data
print("Data shape:", data.shape)
print(hdul.info())
# the brightest pixel value in the data
max_pixel_value = np.max(data)
print("Brightest Pixel Value:", max_pixel_value)

# the darkest pixel value in the data
min_pixel_value = np.min(data)
print("Darkest Pixel Value:", min_pixel_value)

# the mean pixel value in the data
mean_pixel_value = np.mean(data)
print("Mean Pixel Value:", mean_pixel_value)

# Visualizing the data
plt.figure(figsize=(8,8))
plt.imshow(data, cmap='inferno', origin='lower')
plt.colorbar(label="Pixel Intensity")
plt.title("Messier 12")
plt.show()

# using Logarithmic scaling to enhance the visibility
from matplotlib.colors import LogNorm
plt.figure(figsize=(8,8))
plt.imshow(data, cmap='inferno', origin='lower', norm=LogNorm())
plt.colorbar(label="Pixel Intensity (Log Scale)")
plt.title("Messier 12 (Log Scale)")
plt.show()
```

### 4.2 Background Estimation and Star Detection

```
# Estimating background statistics using sigma clipping
bg_mean, bg_median, bg_std = sigma_clipped_stats(data, sigma=3.0)
print("Background Mean:", bg_mean)
print("Background Median:", bg_median)
print("Background Standard Deviation:", bg_std)

# Detecting stars using the DAOStarFinder Algorithm
from photutils.detection import DAOStarFinder
```

```

daofind = DAOStarFinder(fwhm=3.0, threshold=5.0*bg_std)
stars = daofind(data - bg_median)
for col in stars.colnames:
    if col not in ('id', 'npix'):
        stars[col].info.format = '%.2f'
stars.pprint(max_width=76)

from photutils.aperture import CircularAperture

positions = np.transpose((stars['x_centroid'], stars['y_centroid']))
apertures = CircularAperture(positions, r=7)

# Visualize the detected stars
plt.figure(figsize=(10,10))
plt.imshow(data, cmap='inferno', origin='lower', norm=LogNorm(),
           interpolation='nearest')
apertures.plot(color='green', lw=1.5, alpha=0.7)
plt.colorbar(label="Pixel Intensity (Log Scale)")
plt.title("Detected Stars in Messier 12")
plt.show()

```

### 4.3 Histogram of the flux of the detected stars

```

# Extracting the flux values of the detected sources and plotting a
# histogram to visualize the distribution of fluxes among the detected
# stars
fluxes = stars['flux']

# Histogram
plt.figure(figsize=(10,6))
plt.hist(fluxes, bins=20, color='blue', edgecolor='black', alpha=0.7)

# Labels and title
plt.xlabel('Flux (Brightness)')
plt.ylabel('Number of Stars')
plt.title('Flux Distribution of Detected Stars in Messier 12')
plt.show()

# Convert flux values to logarithmic scale
log_fluxes = np.log10(fluxes)

# Plot histogram
plt.figure(figsize=(10,6))
plt.hist(log_fluxes, bins=30, color='orange', edgecolor='black', alpha=0.7)
plt.xlabel('log10(Flux)')
plt.ylabel('Number of Stars')
plt.title('Log-Flux Distribution of Detected Stars in Messier 12')
plt.show()

```

## 5 References

1. Isabell Hudnall, *Photutils.detection — Detecting Stars with DAOSTarFinder*, YouTube video tutorial. Available at: [https://www.youtube.com/watch?v=vxMoG2K\\_n5o](https://www.youtube.com/watch?v=vxMoG2K_n5o)
2. Hubble Space Telescope (HST) Archive, *NASA/ESA Hubble Space Telescope Data Archive*. Available at: <https://archive.stsci.edu/hst/>
3. Sanjeev K., *M12 Globular Cluster Analysis using Python*, GitHub repository. Available at: [https://github.com/arsanjeevk/M12\\_Globular\\_Cluster](https://github.com/arsanjeevk/M12_Globular_Cluster)